Disc brake rotor CAD model and CAE thermo-mechanical analysis

Scenario

Vehicle mass = 1500 kg

Initial speed = 100 km/h = 27.78 m/s

Final speed = 0 m/s [vehicle is stopping]

Braking time = 4 s

Braking bias = front brakes(70%) AND rear brakes(30%)

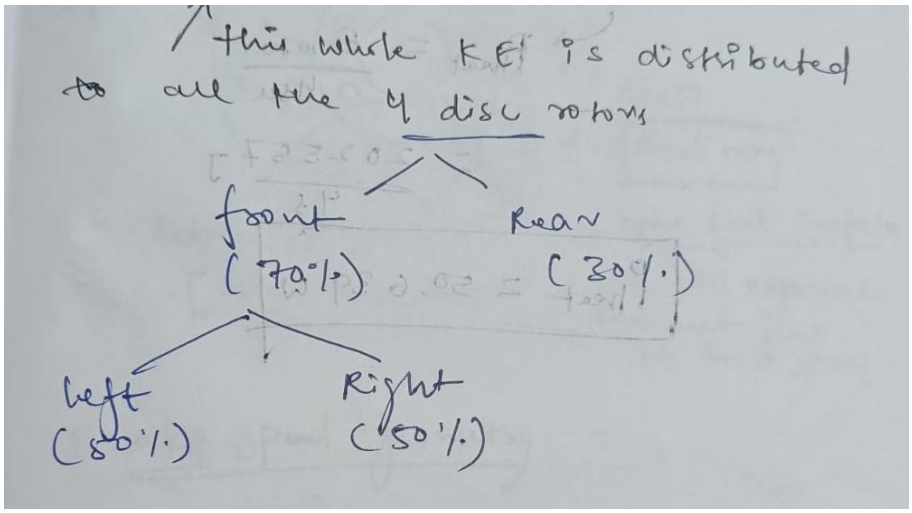
We are considering one of the front brake rotors

While stopping all the kinetic energy of the vehicle is transferred to the brake rotor which dissipates that energy in the form of heat.

KE = $\frac{1}{2}mv^2$ = $\frac{1}{2}(1500\ kg)(27.78^2)$

KE = 578,734 J

This whole KE is distributed to all the 4 wheel/disc rotors.



Energy transferred to one of the front wheels is

E_{Rotor} $= \frac{1}{2}(Energy\ on\ the\ front\ two\ wheels)$

$= \frac{1}{2}(70\%\ of\ Total\ KE\ )$

$= \frac{1}{2}(\frac{70}{100} \times 578{,}734)$

E_{Rotor} = 202,557 J

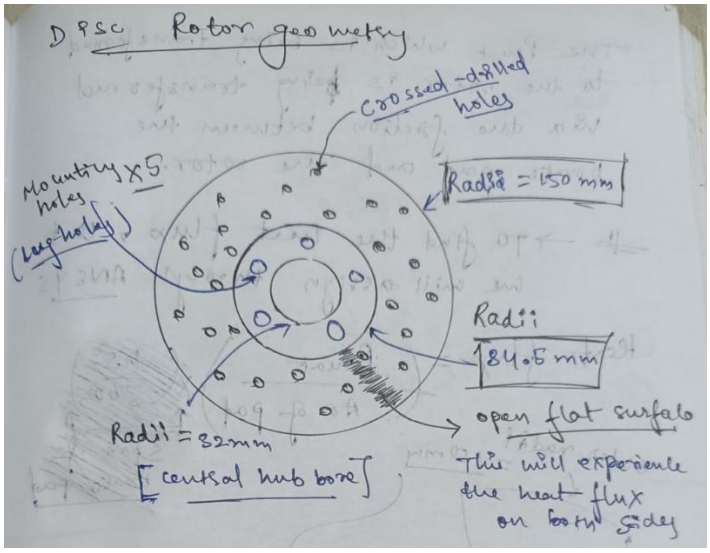
The energy E_{Rotor} is experienced by the rotor disc for 4 s(as we have assumed).

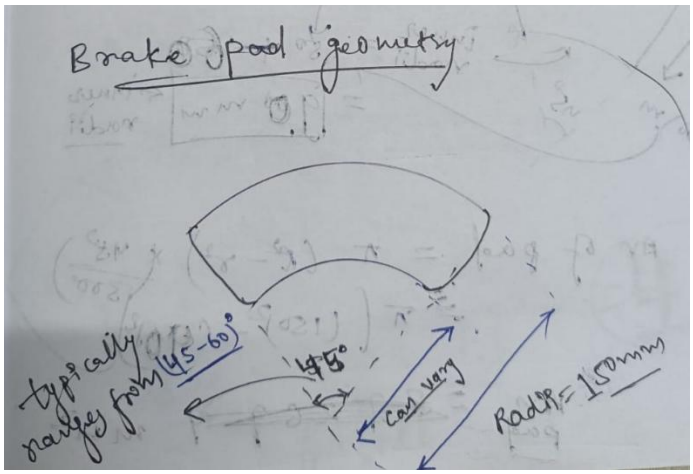
So, the total heat generated through that time is :

$P_{Heat} = \frac{E_{Rotor}}{4s}$

$= \frac{202{,}557\ \text{W}}{4\ s}$

= 50,639 W





The P_{Heat} is transferred to the rotor through friction between the rotor and the brake pads.

Heat Flux

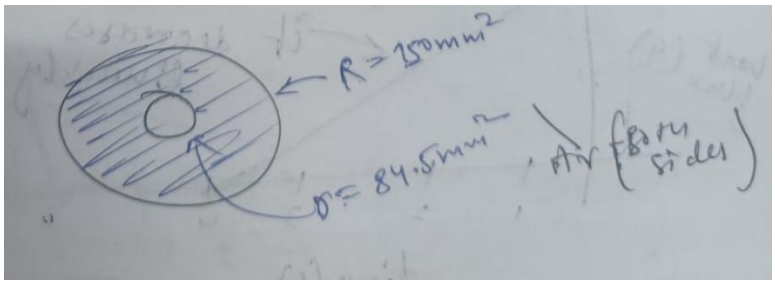
The heat flux(q) is

$$q = \frac{P_{Heat}}{A_{Total}}$$

A_{Total} = Area onto which this whole P_{Heat} will be transferred.

It includes the area of both the sides of the rotor disc.

We are considering the transient thermal so , we will distribute all the heat onto the flat surfaces on both the sides(360° $smeared$).



$$A_{Total} = 2(\pi(R^2 - r^2))$$

$$= 2(\pi(150^2 - 84.5^2))$$

$$= 96508\ mm^2$$

Now , we can calculate the heat flux(q)

$$q = \frac{P_{Heat}}{A_{Total}}$$

$$= \frac{50639\ W}{96508\ mm^2}$$

$q = 0.525\ W/mm^2$.

Throughout the 4s brake, the heat flux will vary like this.



Tabular Data

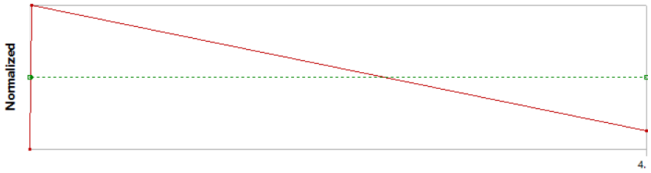
	Steps	Time [s]	Heat Flux [W/mm²]
1	1	0.	0.
2	1	1.e-002	0.54
3	1	4.	0.
*			

Convection

Similarly for convection, we will add the film coefficient(*h*) in linearly decreasing manner.

And the max value of *h* is 80 $W/m^2.°C$ or $8 \times 10^5 W/mm^2.°C$ (when the vehicle is at full speed, the air moves faster around the surfaces creating a forced convection conditions).

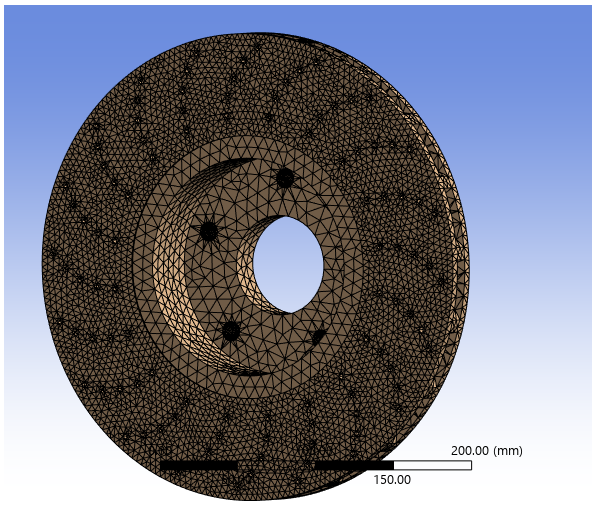
The min value of *h* is 10 $W/m^2.°C$ or $1 \times 10^5 W/mm^2.°C$ (when the vehicle is at rest the air moves slow creating the natural convection).



Tabular Data

	Steps	Time [s]	Convection Coefficient [W/mm²·°C]	Temperature [°C]
1	1	0.	0.	= 25.
2	1	1.e-002	8.e-005	25.
3	1	4.	1.e-005	= 25.

Meshing



Material

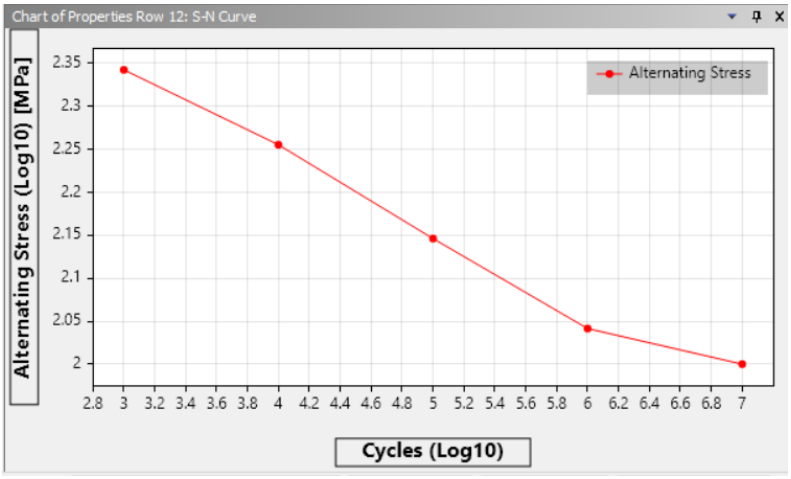
Gray cast iron

Properties of Outline Row 3: Gray Cast Iron

	A	B	C	D	E
1	Property	Value	Unit		
2	Material Field Variables	Table			
3	Density	7200	kg m^-3		
4	Isotropic Secant Coefficient of Thermal Expansion				
5	Coefficient of Thermal Expansion	1.1E-05	C^-1		
6	Isotropic Elasticity				
7	Derive from	Young's Modulu...			
8	Young's Modulus	1.1E+11	Pa		
9	Poisson's Ratio	0.28			
10	Bulk Modulus	8.3333E+10	Pa		
11	Shear Modulus	4.2969E+10	Pa		

12	S-N Curve	Tabular	
13	Interpolation	Log-Log	
14	Scale	1	
15	Offset	0	MPa
16	Tensile Yield Strength	0	Pa
17	Compressive Yield Strength	0	Pa
18	Tensile Ultimate Strength	2.4E+08	Pa
19	Compressive Ultimate Strength	8.2E+08	Pa
20	Isotropic Thermal Conductivity	52	W m^-1 C^-1
21	Specific Heat Constant Pressure, C_p	447	J kg^-1 C^-1
22	Isotropic Resistivity	9.6E-08	ohm m
23	Isotropic Relative Permeability	10000	

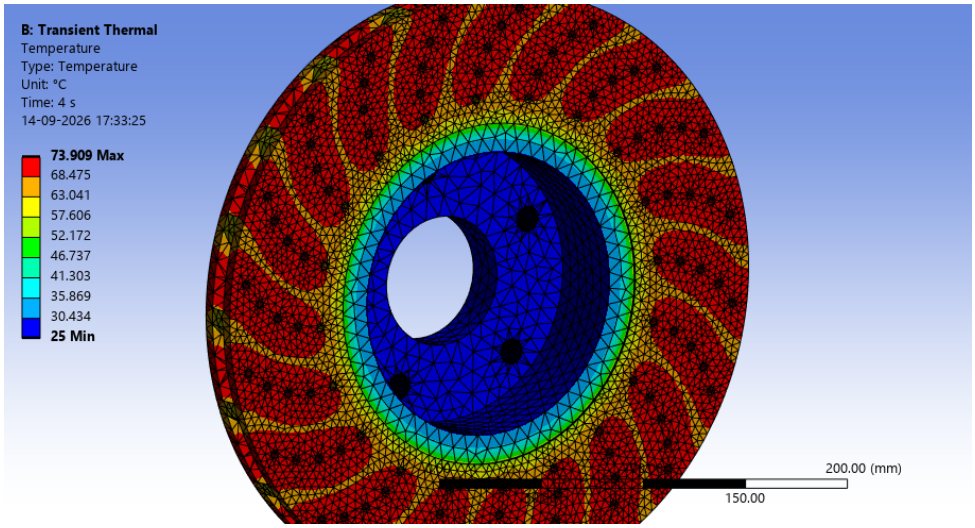
S v/s N Curve

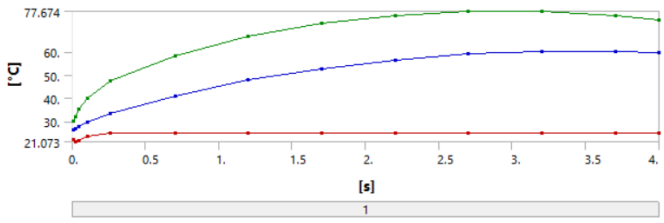


Boundary conditions

Results

Temperature

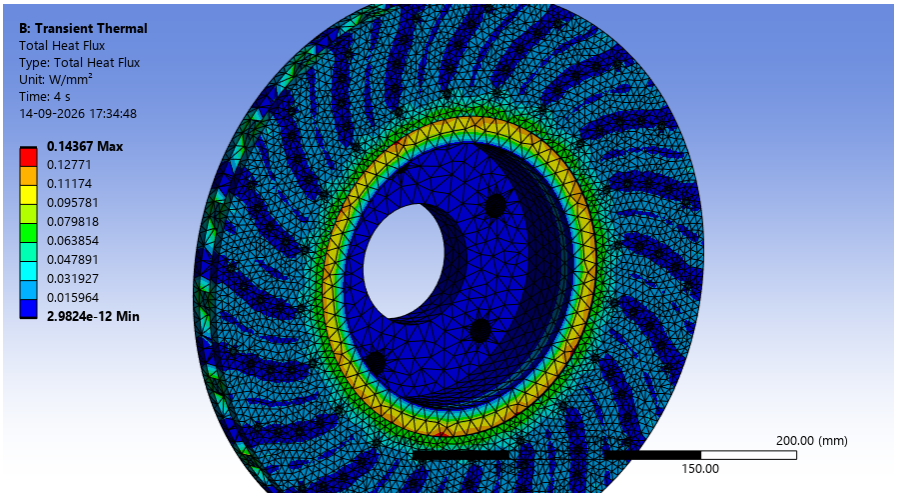


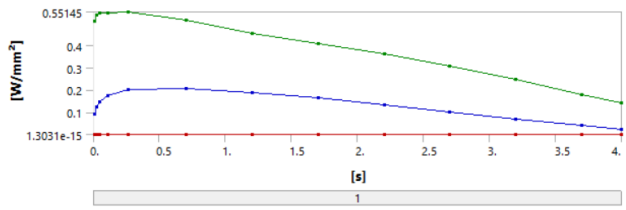


Tabular Data

	Time [s]	Minimum [°C]	Maximum [°C]	Average [°C]
1	1.e-002	21.951	30.058	26.029
2	2.e-002	21.073	31.947	26.585
3	4.4581e-002	21.439	35.139	27.639
4	0.10626	23.666	40.083	29.638
5	0.2605	24.937	47.474	33.395
6	0.70083	25.	58.467	41.088
7	1.2008	25.	66.668	47.776
8	1.7008	25.	72.39	52.91
9	2.2008	25.	75.963	56.655
10	2.7008	25.	77.674	59.081
11	3.2008	25.	77.572	60.226
12	3.7008	25.	75.679	60.115
13	4.	25.	73.909	59.603

Total heat flux





Tabular Data

	Time [s]	Minimum [W/mm²]	Maximum [W/mm²]	Average [W/mm²]
1	1.e-002	7.4491e-011	0.51114	9.3189e-002
2	2.e-002	8.2217e-011	0.53723	0.12206
3	4.4581e-002	2.1838e-010	0.54684	0.14826
4	0.10626	2.1996e-011	0.54846	0.17398
5	0.2605	6.8208e-013	0.55145	0.20156
6	0.70083	1.0108e-014	0.51598	0.2077
7	1.2008	1.3031e-015	0.45688	0.19044
8	1.7008	1.2336e-014	0.41083	0.16434
9	2.2008	9.7884e-014	0.36418	0.13459
10	2.7008	3.201e-013	0.30846	0.10325
11	3.2008	7.7341e-013	0.24613	7.1199e-002
12	3.7008	3.1455e-012	0.1807	3.9329e-002
13	4.	2.9824e-012	0.14367	2.2672e-002

Static Structural analysis

Further to know the effects of the actual condition of braking on the structure of the disc rotor we will use static structure